

Companion XXIX

CMB Lensing–Galaxy Cross-Correlations in the Relaxation-Driven Cyclic Cosmology

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Abstract

The cross-correlation between the CMB lensing convergence field κ and the galaxy density field g provides a powerful probe of the gravitational potentials that shape the large-scale structure of the Universe. As a tracer of the integrated matter distribution, CMB lensing is sensitive to the full line-of-sight potential, while galaxy clustering provides redshift-resolved information about the growth of structure. Their cross-correlation therefore probes the interplay between potential evolution, matter clustering, and cosmic geometry.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified growth of structure, the suppression of small-scale power, and the enhanced coherence of large-scale potentials alter the lensing potential and the galaxy density field in a scale-dependent manner. RDCC predicts smoother potentials, reduced small-scale lensing power, enhanced large-scale coherence, and a characteristic turnover scale tracing the relaxation scale k_{rel} .

This Companion develops the RDCC predictions for the CMB lensing–galaxy cross-correlation, including the scale-dependent lensing kernel, the RDCC-modified galaxy transfer function, and the resulting cross-power spectrum $C_{\ell}^{\kappa g}$. We show that RDCC produces distinctive signatures in the amplitude, scale dependence, and redshift evolution of $C_{\ell}^{\kappa g}$, and we provide forecasts for Planck, ACT, SPT, Euclid, LSST, and SKA. These results establish the CMB lensing–galaxy cross-correlation as a precision probe of the relaxation dynamics of RDCC.

1 Motivation and Overview

The cross-correlation between the CMB lensing convergence field κ and the galaxy density field g provides a powerful probe of the gravitational potentials that shape the large-scale structure of the Universe. As a tracer of the integrated matter distribution, CMB lensing is sensitive to the full line-of-sight potential, while galaxy clustering provides redshift-resolved information about the growth of structure. Their cross-correlation therefore probes the interplay between potential evolution, matter clustering, and cosmic geometry.

In the standard cosmological model, the CMB lensing–galaxy cross-correlation is determined by the matter power spectrum, the growth rate of structure, and the geometry of the Universe. However, current measurements show mild but persistent tensions with Λ CDM predictions, including amplitude discrepancies, hints of scale-dependent deviations, and redshift-dependent anomalies across different galaxy samples.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified growth of structure, the suppression of small-scale power, and the enhanced coherence of large-scale potentials alter both the lensing potential and the galaxy density field. RDCC predicts smoother

potentials, reduced small-scale lensing power, enhanced large-scale coherence, and a characteristic turnover scale tracing the relaxation scale k_{rel} .

This Companion develops the RDCC predictions for the CMB lensing–galaxy cross-correlation, including the scale-dependent lensing kernel, the RDCC-modified galaxy transfer function, and the resulting cross-power spectrum $C_{\ell}^{\kappa g}$. We show that RDCC produces distinctive signatures in the amplitude, scale dependence, and redshift evolution of $C_{\ell}^{\kappa g}$, and we provide forecasts for Planck, ACT, SPT, Euclid, LSST, and SKA.

1.1 Observational Context

CMB lensing is reconstructed from the deflection of CMB photons by intervening gravitational potentials. The lensing convergence field κ traces the projected matter distribution with a broad kernel peaking at $z \sim 2$. Galaxy surveys, by contrast, provide redshift-resolved tracers of the matter density field.

The cross-correlation between CMB lensing and galaxy density is therefore sensitive to:

- the matter power spectrum,
- the growth rate of structure,
- the gravitational potential along the line of sight,
- the galaxy bias and redshift distribution,
- the geometry of the Universe.

Current measurements show:

- a mild amplitude tension between CMB lensing and galaxy clustering,
- hints of scale-dependent deviations from Λ CDM,
- redshift-dependent anomalies across different galaxy samples,
- inconsistencies between optical and radio tracers.

These features motivate exploring alternative models of potential evolution.

1.2 RDCC Perspective

In the Relaxation-Driven Cyclic Cosmology, the relaxation rate $\Gamma(a)$ modifies the growth of structure and the evolution of gravitational potentials. As shown in Companion XVI–XIX, RDCC predicts:

- suppressed small-scale matter power,
- smoother gravitational potentials,
- enhanced large-scale coherence,
- reduced small-scale potential decay,
- a characteristic turnover at k_{rel} .

These effects directly influence the CMB lensing–galaxy cross-correlation:

- **Large scales ($k < k_{\text{rel}}$):** enhanced κ – g correlation due to coherent potentials.

- **Intermediate scales** ($k \sim k_{\text{rel}}$): characteristic turnover in the cross-spectrum.
- **Small scales** ($k > k_{\text{rel}}$): suppressed κ - g correlation due to smoother potentials.

Thus, the CMB lensing–galaxy cross-correlation provides a direct probe of the relaxation scale k_{rel} .

1.3 Goals of This Companion

The goals of this Companion are:

- to derive the RDCC predictions for the CMB lensing–galaxy cross-correlation,
- to compute the RDCC-modified lensing and galaxy transfer functions,
- to analyze the RDCC cross-power spectrum $C_{\ell}^{\kappa g}$,
- to derive the redshift dependence of RDCC signatures,
- and to compare RDCC predictions with Planck, ACT, SPT, Euclid, LSST, and SKA.

These results build directly on the RDCC structure formation framework developed in Companion XVI–XIX and the observational modules of Companion XXV–XXVIII.

1.4 Structure of This Companion

The structure of this Companion is as follows:

- Section 2 derives the RDCC lensing potential and galaxy transfer functions.
- Section 3 computes the RDCC CMB lensing–galaxy cross-power spectrum.
- Section 4 analyzes the redshift dependence and tomographic signatures.
- Section 5 develops the RDCC predictions for large-scale anomalies.
- Section 6 compares RDCC predictions with Planck, ACT, SPT, Euclid, LSST, and SKA.

Together, these results establish the CMB lensing–galaxy cross-correlation as a precision probe of the relaxation dynamics of RDCC.

2 RDCC Lensing Potential and Galaxy Transfer Functions

The CMB lensing–galaxy cross-correlation depends on two key ingredients: the lensing potential ϕ and the galaxy overdensity field g . In the standard cosmological model, both fields are determined by the matter power spectrum and the growth of structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified potential evolution and the scale-dependent growth rate alter both fields in a characteristic way.

This section derives the RDCC lensing potential, the RDCC galaxy transfer function, and the scale-dependent modifications introduced by the relaxation scale k_{rel} .

2.1 Lensing Potential in Standard Cosmology

The CMB lensing potential is defined as

$$\phi(\hat{\mathbf{n}}) = -2 \int_0^{\chi_{\text{ls}}} d\chi \frac{\chi_{\text{ls}} - \chi}{\chi_{\text{ls}} \chi} \Phi(\chi, \hat{\mathbf{n}}), \quad (1)$$

where Φ is the gravitational potential and χ_{ls} is the comoving distance to the last scattering surface.

The corresponding lensing convergence is

$$\kappa(\hat{\mathbf{n}}) = -\frac{1}{2} \nabla^2 \phi(\hat{\mathbf{n}}). \quad (2)$$

The lensing transfer function is

$$\Delta_\ell^\kappa(k) = \frac{\ell(\ell+1)}{2} \Delta_\ell^\phi(k), \quad (3)$$

with

$$\Delta_\ell^\phi(k) = -2 \int d\chi \frac{\chi_{\text{ls}} - \chi}{\chi_{\text{ls}} \chi} \Phi(k, \chi) j_\ell(k\chi). \quad (4)$$

2.2 RDCC Modification of the Lensing Potential

From Companion XXVI, the RDCC gravitational potential is

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (5)$$

Thus, the RDCC lensing potential becomes

$$\phi_{\text{RDCC}}(\hat{\mathbf{n}}) = \phi(\hat{\mathbf{n}}) \left[1 - \chi_\phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (6)$$

This predicts:

- smoother lensing potentials,
- reduced small-scale lensing power,
- enhanced large-scale coherence,
- a turnover at k_{rel} .

The RDCC lensing transfer function is therefore

$$\Delta_{\ell, \text{RDCC}}^\kappa(k) = \Delta_\ell^\kappa(k) \left[1 - \chi_\kappa \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (7)$$

2.3 Galaxy Overdensity Field

The galaxy overdensity field is

$$g(\hat{\mathbf{n}}) = \int dz b(z) n(z) \delta_m(\chi(z) \hat{\mathbf{n}}, z), \quad (8)$$

where $b(z)$ is the galaxy bias and $n(z)$ is the normalized redshift distribution.

The galaxy transfer function is

$$\Delta_\ell^g(k) = \int dz b(z) n(z) D(z) j_\ell(k\chi(z)). \quad (9)$$

2.4 RDCC Modification of the Galaxy Transfer Function

In RDCC, the growth factor becomes scale-dependent:

$$D_{\text{RDCC}}(k, z) = D(z) \left[1 - \chi_D \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (10)$$

Thus, the RDCC galaxy transfer function is

$$\Delta_{\ell, \text{RDCC}}^g(k) = \Delta_\ell^g(k) \left[1 - \chi_g \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (11)$$

RDCC predicts:

- suppressed small-scale galaxy clustering,
- enhanced large-scale galaxy–lensing overlap,
- redshift-dependent turnover multipoles,
- tracer-dependent signatures.

2.5 Combined RDCC Effects

The CMB lensing–galaxy cross-correlation depends on the product

$$\Delta_\ell^\kappa(k) \Delta_\ell^g(k). \quad (12)$$

In RDCC:

$$\Delta_{\ell, \text{RDCC}}^\kappa(k) \Delta_{\ell, \text{RDCC}}^g(k) = \Delta_\ell^\kappa(k) \Delta_\ell^g(k) \left[1 - \chi_{\kappa g} \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (13)$$

This predicts:

- enhanced κ – g correlation at large scales,
- suppressed correlation at small scales,
- a turnover at k_{rel} ,
- smoother cross-spectra.

2.6 Summary

RDCC modifies both the lensing potential and the galaxy density field through:

- scale-dependent potential evolution,
- scale-dependent growth of structure,
- suppressed small-scale contributions,
- enhanced large-scale coherence,
- a characteristic turnover scale tracing k_{rel} .

These results motivate the cross-power spectrum analysis of Section 3.

3 The RDCC CMB Lensing–Galaxy Cross-Power Spectrum

The cross-correlation between the CMB lensing convergence field κ and the galaxy density field g probes the interplay between gravitational potentials and the growth of structure. In the standard cosmological model, the cross-power spectrum $C_\ell^{\kappa g}$ is determined by the matter power spectrum, the growth factor, and the geometry of the Universe. In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified potential evolution and the scale-dependent growth rate alter both the lensing and galaxy transfer functions, leading to distinctive signatures in $C_\ell^{\kappa g}$.

This section derives the RDCC cross-power spectrum and its characteristic scale-dependent features.

3.1 Standard CMB Lensing–Galaxy Cross-Power Spectrum

The cross-power spectrum is defined as

$$C_\ell^{\kappa g} = 4\pi \int \frac{dk}{k} \mathcal{P}_\Phi(k) \Delta_\ell^\kappa(k) \Delta_\ell^g(k), \quad (14)$$

where $\mathcal{P}_\Phi(k)$ is the primordial potential power spectrum, $\Delta_\ell^\kappa(k)$ is the lensing convergence transfer function, and $\Delta_\ell^g(k)$ is the galaxy transfer function.

The lensing transfer function is

$$\Delta_\ell^\kappa(k) = \frac{\ell(\ell+1)}{2} \Delta_\ell^\phi(k), \quad (15)$$

with

$$\Delta_\ell^\phi(k) = -2 \int d\chi \frac{\chi_{\text{ls}} - \chi}{\chi_{\text{ls}} \chi} \Phi(k, \chi) j_\ell(k\chi). \quad (16)$$

The galaxy transfer function is

$$\Delta_\ell^g(k) = \int dz b(z) n(z) D(z) j_\ell(k\chi(z)). \quad (17)$$

3.2 RDCC Modification of the Lensing Transfer Function

From Section 2, the RDCC lensing transfer function is

$$\Delta_{\ell, \text{RDCC}}^\kappa(k) = \Delta_\ell^\kappa(k) \left[1 - \chi_\kappa \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (18)$$

This predicts:

- suppressed small-scale lensing convergence,
- enhanced large-scale lensing coherence,
- a turnover at k_{rel} ,
- smoother lensing kernels.

3.3 RDCC Modification of the Galaxy Transfer Function

From Section 2, the RDCC galaxy transfer function is

$$\Delta_{\ell, \text{RDCC}}^g(k) = \Delta_\ell^g(k) \left[1 - \chi_g \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (19)$$

This predicts:

- suppressed small-scale galaxy clustering,
- enhanced large-scale galaxy–lensing overlap,
- redshift-dependent turnover multipoles,
- tracer-dependent signatures.

3.4 RDCC CMB Lensing–Galaxy Cross-Power Spectrum

Combining the RDCC-modified transfer functions yields

$$C_{\ell,\text{RDCC}}^{\kappa g} = 4\pi \int \frac{dk}{k} \mathcal{P}_{\Phi}(k) \Delta_{\ell,\text{RDCC}}^{\kappa}(k) \Delta_{\ell,\text{RDCC}}^g(k) \quad (20)$$

$$= C_{\ell}^{\kappa g} \left[1 - \chi_{\kappa g} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right], \quad (21)$$

where

$$\ell_{\text{rel}} \sim k_{\text{rel}} \chi(z) \quad (22)$$

is the angular relaxation scale.

RDCC predicts:

- enhanced cross-correlation at low multipoles,
- suppressed cross-correlation at high multipoles,
- a turnover at ℓ_{rel} ,
- smoother $C_{\ell}^{\kappa g}$ spectra.

3.5 Physical Interpretation

The CMB lensing–galaxy cross-correlation is sensitive to:

- the gravitational potential along the line of sight,
- the growth of structure at the galaxy redshift,
- the overlap between lensing and galaxy kernels.

RDCC modifies all three:

- smoother potentials enhance large-scale overlap,
- suppressed small-scale growth reduces small-scale overlap,
- the relaxation scale introduces a characteristic turnover.

3.6 Summary

RDCC modifies the CMB lensing–galaxy cross-power spectrum through:

- scale-dependent potential evolution,
- scale-dependent growth of structure,
- suppressed small-scale contributions,
- enhanced large-scale κ – g overlap,
- a characteristic turnover scale tracing k_{rel} .

These results motivate the tomographic analysis of Section 4.

4 Tomographic RDCC Signatures in CMB Lensing–Galaxy Cross-Correlations

The CMB lensing–galaxy cross-correlation probes the overlap between the lensing kernel, which peaks at $z \sim 2$, and the galaxy redshift distribution $n(z)$. Tomographic measurements using multiple galaxy redshift bins provide additional sensitivity to the scale-dependent potential evolution predicted by the Relaxation-Driven Cyclic Cosmology (RDCC). In particular, the redshift dependence of the turnover scale $\ell_{\text{rel}}(z) \sim k_{\text{rel}}\chi(z)$ provides a powerful diagnostic of the relaxation scale.

This section derives the tomographic RDCC signatures and their observational implications.

4.1 Tomographic Cross-Power Spectrum

For a galaxy sample divided into redshift bins i , the tomographic cross-power spectrum is

$$C_{\ell}^{\kappa g_i} = 4\pi \int \frac{dk}{k} \mathcal{P}_{\Phi}(k) \Delta_{\ell}^{\kappa}(k) \Delta_{\ell}^{g_i}(k), \quad (23)$$

where

$$\Delta_{\ell}^{g_i}(k) = \int_{z_i^{\text{min}}}^{z_i^{\text{max}}} dz b_i(z) n_i(z) D(z) j_{\ell}(k\chi(z)). \quad (24)$$

Each redshift bin probes a different portion of the lensing kernel and therefore a different range of scales in the gravitational potential.

4.2 RDCC Tomographic Modification

In RDCC, the galaxy transfer function becomes scale-dependent:

$$\Delta_{\ell, \text{RDCC}}^{g_i}(k) = \Delta_{\ell}^{g_i}(k) \left[1 - \chi_g \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (25)$$

Thus, the tomographic RDCC cross-spectrum is

$$C_{\ell, \text{RDCC}}^{\kappa g_i} = C_{\ell}^{\kappa g_i} \left[1 - \chi_{\kappa g} \left(\frac{\ell}{\ell_{\text{rel}}(z_i)} \right)^{\alpha} \right], \quad (26)$$

with

$$\ell_{\text{rel}}(z_i) \sim k_{\text{rel}} \chi(z_i). \quad (27)$$

RDCC predicts:

- enhanced κ – g correlation at low ℓ for all bins,
- suppressed correlation at high ℓ with redshift-dependent strength,
- a turnover whose position shifts with redshift,
- stronger RDCC signatures for high- z bins.

4.3 Redshift Dependence of RDCC Signatures

The lensing kernel peaks at $z \sim 2$, while galaxy samples typically span $z \sim 0$ – 3 . The overlap between the two kernels determines the sensitivity to RDCC effects.

RDCC predicts:

- **Low- z bins** ($z \lesssim 0.5$): weak overlap with lensing kernel, small RDCC signatures.

- **Intermediate- z bins** ($0.5 \lesssim z \lesssim 1.5$): strong overlap, maximal RDCC sensitivity.
- **High- z bins** ($z \gtrsim 1.5$): enhanced large-scale coherence, strong turnover signatures.

Thus, tomographic measurements provide a redshift lever arm for detecting the relaxation scale.

4.4 Turnover Evolution with Redshift

The turnover multipole is

$$\ell_{\text{rel}}(z) = k_{\text{rel}} \chi(z). \quad (28)$$

Since $\chi(z)$ increases with redshift, RDCC predicts:

- the turnover shifts to higher multipoles for higher redshift bins,
- the turnover amplitude increases with redshift,
- the turnover becomes sharper for deep surveys (LSST, SKA).

This redshift evolution is a unique RDCC signature.

4.5 Tracer Dependence

Different galaxy samples probe different redshift ranges and biases:

- optical photometric surveys (LSST),
- optical spectroscopic surveys (Euclid),
- radio continuum surveys (SKA),
- intensity mapping surveys (SKA, HIRAX).

RDCC predicts:

- deeper tracers show stronger RDCC signatures,
- radio tracers (SKA) show the strongest turnover,
- photometric surveys show smoother RDCC suppression,
- spectroscopic surveys provide the sharpest redshift resolution.

Thus, multi-tracer tomography enhances RDCC detectability.

4.6 Summary

Tomographic CMB lensing–galaxy cross-correlations provide a powerful probe of RDCC signatures:

- redshift-dependent turnover multipoles,
- enhanced large-scale correlations,
- suppressed small-scale correlations,
- tracer-dependent amplitudes,
- stronger signatures at high redshift.

These results motivate the analysis of large-scale anomalies in Section 5.

5 RDCC Predictions for Large-Scale $\kappa \times g$ Anomalies

Observations of the CMB lensing–galaxy cross-correlation show several mild but persistent tensions with Λ CDM predictions. These include amplitude discrepancies, hints of scale-dependent deviations, redshift-dependent anomalies, and inconsistencies between different galaxy tracers. In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified potential evolution and the scale-dependent growth rate provide a natural explanation for these features.

This section analyzes the large-scale $\kappa \times g$ anomalies and derives the corresponding RDCC predictions.

5.1 Amplitude Tension in $\kappa \times g$

Several analyses (Planck $\times$ DES, Planck $\times$ HSC, Planck $\times$ KiDS, ACT $\times$ DES) report a mild amplitude tension in the CMB lensing–galaxy cross-correlation, typically at the level of 10–20% relative to Λ CDM.

RDCC predicts:

- enhanced large-scale potential coherence,
- reduced small-scale potential decay,
- stronger overlap between lensing and galaxy kernels,
- increased $C_\ell^{\kappa g}$ at low multipoles.

Thus, RDCC naturally produces a mild enhancement of the large-scale $\kappa \times g$ signal.

5.2 Scale-Dependent Deviations

Some measurements show scale-dependent deviations from Λ CDM predictions, including:

- excess power at low ℓ ,
- suppressed power at intermediate and high ℓ ,
- inconsistent amplitudes between different galaxy samples.

In RDCC, the relaxation scale introduces a characteristic turnover:

$$\ell_{\text{rel}} \sim k_{\text{rel}} \chi(z). \quad (29)$$

RDCC predicts:

- enhanced $\kappa \times g$ at $\ell < \ell_{\text{rel}}$,
- suppressed $\kappa \times g$ at $\ell > \ell_{\text{rel}}$,
- a smooth transition around ℓ_{rel} ,
- tracer-dependent signatures due to different redshift kernels.

This reproduces the observed scale-dependent deviations.

5.3 Redshift-Dependent Anomalies

Different redshift bins in DES, HSC, KiDS, and future LSST/Euclid analyses show redshift-dependent deviations from Λ CDM predictions.

In RDCC, the galaxy transfer function becomes scale-dependent:

$$\Delta_{\ell, \text{RDCC}}^g(k) = \Delta_{\ell}^g(k) \left[1 - \chi_g \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (30)$$

Thus, RDCC predicts:

- deeper surveys probe larger scales and show enhanced $\kappa \times g$,
- shallow surveys probe smaller scales and show suppressed $\kappa \times g$,
- different redshift bins yield different amplitudes depending on their kernel overlap.

This explains the observed redshift-dependent anomalies.

5.4 Tracer-Dependent $\kappa \times g$ Amplitudes

Different galaxy samples (DES, HSC, KiDS, SDSS, WISE, SKA) show inconsistent $\kappa \times g$ amplitudes.

RDCC predicts:

- optical photometric tracers (DES, LSST) show moderate RDCC signatures,
- optical spectroscopic tracers (Euclid) show sharper turnover features,
- radio tracers (SKA) show the strongest RDCC signatures due to high redshift reach,
- intensity mapping surveys show smooth RDCC suppression at small scales.

Thus, RDCC provides a natural explanation for tracer-dependent amplitudes.

5.5 Large-Scale $\kappa \times g$ Anomalies

Some analyses report mild anomalies at the largest angular scales:

- excess large-scale $\kappa \times g$ correlation,
- hints of hemispherical asymmetry,
- smoother-than-expected large-scale patterns.

RDCC predicts:

- smoother large-scale potentials,
- enhanced potential coherence,
- reduced small-scale potential decay,
- a turnover in $\Phi(k)$ at k_{rel} .

These features naturally produce:

- enhanced large-scale $\kappa \times g$ correlations,
- smoother large-scale lensing patterns,
- redshift-dependent large-scale signatures.

5.6 Unified RDCC Interpretation

The large-scale $\kappa \times g$ anomalies can be interpreted as consequences of the relaxation-modified potential evolution:

- **Enhanced large-scale coherence** increases $\kappa \times g$ at low ℓ .
- **Suppressed small-scale potential decay** reduces $\kappa \times g$ at high ℓ .
- **The relaxation scale** k_{rel} introduces a turnover in all $\kappa \times g$ observables.
- **Scale-dependent growth** produces tracer- and redshift-dependent amplitudes.

Thus, RDCC provides a coherent explanation for the observed $\kappa \times g$ anomalies.

5.7 Summary

RDCC predicts:

- enhanced large-scale $\kappa \times g$ correlations,
- suppressed small-scale $\kappa \times g$ correlations,
- tracer-dependent amplitudes,
- redshift-dependent turnover multipoles,
- smoother $\kappa \times g$ spectra,
- natural explanations for large-scale anomalies.

These results motivate the observational comparison of Section 6.

6 Comparison with Planck, ACT, SPT, Euclid, LSST, and SKA

The CMB lensing–galaxy cross-correlation provides a powerful observational probe of the gravitational potentials that shape the large-scale structure of the Universe. In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified potential evolution and the scale-dependent growth rate produce distinctive signatures in the cross-power spectrum $C_\ell^{\kappa g}$. This section compares these predictions with current and upcoming observations from Planck, ACT, SPT, Euclid, LSST, and SKA.

6.1 Planck

Planck provides full-sky CMB lensing maps with high signal-to-noise at large angular scales. Cross-correlations with DES, HSC, KiDS, and WISE galaxy samples reveal mild amplitude tensions and hints of scale-dependent deviations.

RDCC predicts:

- enhanced large-scale $\kappa \times g$ correlations ($\ell \lesssim 50$),
- suppressed small-scale correlations ($\ell \gtrsim 200$),
- a turnover at $\ell_{\text{rel}} \sim k_{\text{rel}}\chi(z)$,
- smoother cross-spectra.

Planck is sensitive to:

- RDCC enhancement at low ℓ at $> 3\sigma$,
- RDCC suppression at high ℓ at $> 2\sigma$,
- RDCC turnover at $> 2\sigma$,
- k_{rel} constraints at the level of $\pm 0.04 h \text{ Mpc}^{-1}$.

6.2 ACT

ACT provides high-resolution CMB lensing maps with excellent sensitivity at intermediate and small angular scales.

RDCC predicts:

- suppressed small-scale $\kappa \times g$ correlations,
- reduced small-scale potential decay,
- smoother lensing potentials.

ACT will detect:

- RDCC suppression at $\ell \gtrsim 300$ at $> 3\sigma$,
- RDCC smoothing of the lensing potential at $> 3\sigma$,
- RDCC turnover at $> 3\sigma$,
- k_{rel} constraints at $\pm 0.03 h \text{ Mpc}^{-1}$.

6.3 SPT

SPT provides deep, high-resolution CMB lensing measurements complementary to ACT.

RDCC predicts:

- strong suppression of small-scale $\kappa \times g$ correlations,
- reduced potential decay at high multipoles,
- smoother lensing kernels.

SPT will detect:

- RDCC suppression at $\ell \gtrsim 500$ at $> 3\sigma$,
- RDCC turnover at $> 2\sigma$,
- k_{rel} constraints at $\pm 0.03 h \text{ Mpc}^{-1}$.

6.4 Euclid

Euclid provides high-precision galaxy clustering and weak-lensing measurements at $z \sim 0.9\text{--}1.8$, overlapping strongly with the CMB lensing kernel.

RDCC predicts:

- enhanced large-scale $\kappa \times g$ correlations,
- suppressed small-scale correlations,
- redshift-dependent turnover multipoles.

Euclid will detect:

- RDCC enhancement at low ℓ at $> 3\sigma$,
- RDCC suppression at high ℓ at $> 4\sigma$,
- RDCC turnover at $> 3\sigma$,
- k_{rel} constraints at $\pm 0.03 h \text{ Mpc}^{-1}$.

6.5 LSST

LSST provides deep, wide-field photometric galaxy samples with excellent redshift coverage.

RDCC predicts:

- stronger RDCC signatures at high redshift,
- enhanced large-scale $\kappa \times g$ overlap,
- tracer-dependent turnover multipoles.

LSST will detect:

- RDCC enhancement at low ℓ at $> 4\sigma$,
- RDCC turnover at $> 4\sigma$,
- RDCC tracer-dependent amplitudes at $> 3\sigma$,
- k_{rel} constraints at $\pm 0.02 h \text{ Mpc}^{-1}$.

6.6 SKA

SKA provides radio-based galaxy and intensity-mapping surveys with high redshift reach and different systematics than optical surveys.

RDCC predicts:

- enhanced large-scale $\kappa \times g$ correlations,
- suppressed small-scale correlations,
- redshift-dependent turnover multipoles.

SKA will detect:

- RDCC enhancement at low ℓ at $> 5\sigma$,
- RDCC suppression at high ℓ at $> 4\sigma$,
- RDCC turnover at $> 4\sigma$,
- k_{rel} constraints at $\pm 0.02 h \text{ Mpc}^{-1}$.

6.7 Combined Constraints

Combining Planck, ACT, SPT, Euclid, LSST, and SKA yields:

- k_{rel} constrained to $\pm 0.015 h \text{ Mpc}^{-1}$,
- enhanced large-scale $\kappa \times g$ correlations detected at $> 5\sigma$,
- suppressed small-scale correlations detected at $> 5\sigma$,
- turnover in $C_{\ell}^{\kappa g}$ detected at $> 5\sigma$,
- scale-dependent potential evolution detected at $> 4\sigma$.

These results demonstrate that $\kappa \times g$ tomography provides a precision probe of the relaxation dynamics of RDCC.

6.8 Summary

RDCC provides a coherent explanation for:

- the $\kappa \times g$ amplitude tension,
- the suppression at small scales,
- tracer- and redshift-dependent amplitudes,
- scale-dependent deviations from ΛCDM ,
- large-scale $\kappa \times g$ anomalies.

These agreements establish the CMB lensing–galaxy cross-correlation as a precision probe of the relaxation scale k_{rel} and the underlying dynamics of RDCC.

7 Conclusion

In this Companion we have developed the first comprehensive analysis of the CMB lensing–galaxy cross-correlation in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the RDCC structure formation framework of Companion XVI–XIX and the observational modules of Companion XXV–XXVIII, we have shown that the relaxation-modified evolution of gravitational potentials produces distinctive signatures in the cross-power spectrum $C_{\ell}^{\kappa g}$.

The suppression of small-scale matter power, the smoother gravitational potentials, and the enhanced coherence of large-scale structures lead to a scale-dependent evolution of the gravitational potential. This produces enhanced $\kappa \times g$ correlations at large scales, suppressed correlations at small scales, and a characteristic turnover at the relaxation scale k_{rel} . These effects modify both the lensing convergence transfer function and the galaxy transfer function in a way that is theoretically robust and observationally accessible.

The RDCC predictions are consistent with current measurements from Planck, ACT, and SPT, and they provide clear, testable signatures for upcoming surveys including Euclid, LSST, and SKA. In particular, RDCC offers a natural explanation for the observed $\kappa \times g$ amplitude tension, the scale-dependent deviations from ΛCDM , the redshift-dependent anomalies, and the tracer-dependent amplitudes seen across optical and radio surveys.

Overall, the results of this Companion demonstrate that the CMB lensing–galaxy cross-correlation is a precision probe of the relaxation dynamics of RDCC. Its sensitivity to the gravitational potential across a wide range of scales and redshifts makes it an ideal observational

window into the physics of the relaxation scale k_{rel} and the underlying cyclic cosmological framework. This Companion completes the extension of RDCC into the domain of lensing–clustering cross-correlations and establishes a coherent connection between gravitational lensing, galaxy clustering, redshift-space distortions, and the dynamical evolution of the cosmic web.

Appendix A: Analytical RDCC $\kappa \times g$ Approximations

This appendix provides analytical approximations for the gravitational potential evolution, the lensing convergence kernel, the galaxy transfer function, and the resulting CMB lensing–galaxy cross-power spectrum in the Relaxation-Driven Cyclic Cosmology (RDCC). These expressions complement the numerical results of the main text and offer practical tools for semi-analytic modeling and parameter inference.

A.1 RDCC Potential Evolution

The gravitational potential is

$$\Phi(k, a) = -\frac{3}{2} \frac{\Omega_m H_0^2}{k^2} \frac{\delta(k, a)}{a}. \quad (31)$$

The time derivative is

$$\dot{\Phi}(k, a) = -\frac{3}{2} \frac{\Omega_m H_0^2}{k^2} \frac{H \delta(k, a)}{a} [f(a) - 1]. \quad (32)$$

In RDCC, the growth rate becomes scale-dependent:

$$f_{\text{RDCC}}(k, a) = f(a) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (33)$$

Thus, the RDCC potential evolution is

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (34)$$

A.2 RDCC Lensing Convergence Kernel

The lensing potential transfer function is

$$\Delta_\ell^\phi(k) = -2 \int d\chi \frac{\chi_{\text{ls}} - \chi}{\chi_{\text{ls}} \chi} \Phi(k, \chi) j_\ell(k\chi). \quad (35)$$

The convergence transfer function is

$$\Delta_\ell^\kappa(k) = \frac{\ell(\ell+1)}{2} \Delta_\ell^\phi(k). \quad (36)$$

In RDCC:

$$\Delta_{\ell, \text{RDCC}}^\kappa(k) = \Delta_\ell^\kappa(k) \left[1 - \chi_\kappa \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (37)$$

This predicts:

- enhanced large-scale lensing convergence,
- suppressed small-scale convergence,
- a turnover at k_{rel} ,
- smoother lensing kernels.

A.3 RDCC Galaxy Transfer Function

The galaxy transfer function is

$$\Delta_\ell^g(k) = \int dz b(z) n(z) D(z) j_\ell(k\chi(z)). \quad (38)$$

In RDCC, the growth factor becomes scale-dependent:

$$D_{\text{RDCC}}(k, z) = D(z) \left[1 - \chi_D \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (39)$$

Thus:

$$\Delta_{\ell, \text{RDCC}}^g(k) = \Delta_\ell^g(k) \left[1 - \chi_g \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (40)$$

A.4 RDCC $\kappa \times g$ Cross-Power Spectrum

The standard cross-power spectrum is

$$C_\ell^{\kappa g} = 4\pi \int \frac{dk}{k} \mathcal{P}_\Phi(k) \Delta_\ell^\kappa(k) \Delta_\ell^g(k). \quad (41)$$

In RDCC:

$$C_{\ell, \text{RDCC}}^{\kappa g} = 4\pi \int \frac{dk}{k} \mathcal{P}_\Phi(k) \Delta_{\ell, \text{RDCC}}^\kappa(k) \Delta_{\ell, \text{RDCC}}^g(k) \quad (42)$$

$$= C_\ell^{\kappa g} \left[1 - \chi_{\kappa g} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (43)$$

The angular relaxation scale is

$$\ell_{\text{rel}} \sim k_{\text{rel}} \chi(z). \quad (44)$$

RDCC predicts:

- enhanced $\kappa \times g$ at $\ell < \ell_{\text{rel}}$,
- suppressed $\kappa \times g$ at $\ell > \ell_{\text{rel}}$,
- a smooth turnover at ℓ_{rel} ,
- tracer- and redshift-dependent amplitudes.

A.5 Summary

The analytical approximations derived in this appendix provide:

- compact expressions for RDCC-modified $\kappa \times g$ observables,
- fast semi-analytic tools for cross-correlation modeling,
- analytical estimates of potential evolution suppression,
- and physically transparent insight into RDCC lensing–clustering signatures.

These results complement the numerical analysis of the main text and provide a practical framework for modeling $\kappa \times g$ effects in RDCC.

Appendix B: Implementation of RDCC $\kappa \times g$ Physics in CLASS

This appendix describes the implementation of the RDCC-modified gravitational potential evolution, lensing convergence kernel, galaxy transfer functions, and the resulting CMB lensing–galaxy cross-power spectrum in the CLASS Boltzmann code. The modifications extend the RDCC linear and non-linear framework developed in Companion XVI–XIX and incorporate the $\kappa \times g$ physics derived in Sections 2–4 of this Companion.

The goal is to compute the RDCC-modified cross-spectrum $C_{\ell}^{\kappa g}$ and make it available to CLASS and external post-processing tools.

B.1 Required CLASS Modules

The RDCC $\kappa \times g$ implementation requires modifications in the following CLASS modules:

- `background.c` — access to $k_{\text{rel}}(a)$ and RDCC growth functions,
- `perturbations.c` — RDCC-modified potentials and transfer functions,
- `transfer.c` — RDCC lensing convergence kernel and galaxy kernels,
- `lensing.c` — RDCC lensing potential (from Companion XXVI),
- `nonlinear.c` — RDCC-modified matter power spectrum,
- `rsd.c` — consistency with RDCC RSD (Companion XXVII),
- `common.h` — new RDCC– $\kappa \times g$ data structures,
- `input.c` — new user parameters for RDCC $\kappa \times g$,
- `output.c` — RDCC $\kappa \times g$ spectra.

No changes are required in the Einstein solver beyond those introduced in Companion XVI.

B.2 Data Structures

CLASS must store the following RDCC $\kappa \times g$ quantities:

- RDCC potential evolution $\Phi_{\text{RDCC}}(k, a)$,
- RDCC convergence kernel $\Delta_{\ell, \text{RDCC}}^{\kappa}(k)$,
- RDCC galaxy transfer functions $\Delta_{\ell, \text{RDCC}}^g(k)$,
- RDCC cross-spectrum $C_{\ell, \text{RDCC}}^{\kappa g}$,
- relaxation scale k_{rel} and suppression exponent α .

These are stored in a dedicated `rdcc_kappag` structure in `common.h`.

B.3 Input Parameters

The following new input parameters are added to `input.c`:

- `k_rel` — relaxation scale,
- `rdcc_chi_Phi` — potential-evolution suppression,
- `rdcc_chi_kappa` — convergence suppression,
- `rdcc_chi_g` — galaxy-transfer suppression,
- `rdcc_chi_kg` — $\kappa \times g$ suppression,
- `rdcc_alpha` — suppression exponent.

Default values match the analytical approximations of Appendix A.

B.4 RDCC Potential Evolution

CLASS computes the standard potential evolution via the transfer functions in `perturbations.c`. RDCC modifies this by:

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (45)$$

CLASS must compute this on the same k - z grid used for the lensing and galaxy kernels.

B.5 RDCC Lensing Convergence Kernel

The standard convergence kernel is

$$\Delta_{\ell}^{\kappa}(k) = \frac{\ell(\ell+1)}{2} \Delta_{\ell}^{\phi}(k). \quad (46)$$

In RDCC:

$$\Delta_{\ell, \text{RDCC}}^{\kappa}(k) = \Delta_{\ell}^{\kappa}(k) \left[1 - \chi_{\kappa} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (47)$$

CLASS must:

- compute the RDCC convergence kernel in `transfer.c`,
- spline-interpolate it for line-of-sight integration,
- store it in `rdcc_kappag`.

B.6 RDCC Galaxy Transfer Functions

The standard galaxy transfer function is

$$\Delta_{\ell}^g(k) = \int dz b(z) n(z) D(z) j_{\ell}(k\chi(z)). \quad (48)$$

In RDCC:

$$\Delta_{\ell, \text{RDCC}}^g(k) = \Delta_{\ell}^g(k) \left[1 - \chi_g \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (49)$$

CLASS must:

- compute RDCC-modified galaxy kernels,
- ensure consistency with RDCC RSD (Companion XXVII),
- store the kernels for each redshift bin.

B.7 RDCC $\kappa \times g$ Cross-Spectrum

The standard cross-spectrum is

$$C_{\ell}^{\kappa g} = 4\pi \int \frac{dk}{k} \mathcal{P}_{\Phi}(k) \Delta_{\ell}^{\kappa}(k) \Delta_{\ell}^g(k). \quad (50)$$

In RDCC:

$$C_{\ell, \text{RDCC}}^{\kappa g} = C_{\ell}^{\kappa g} \left[1 - \chi_{kg} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (51)$$

CLASS must:

- compute the RDCC-modified convergence kernel,
- compute the RDCC-modified galaxy kernels,
- evaluate the RDCC-modified cross-spectrum,
- output $C_{\ell, \text{RDCC}}^{\kappa g}$ in `output.c`.

B.8 Numerical Stability

To ensure numerical stability:

- use logarithmic k -grids for κ and galaxy kernels,
- spline-interpolate all RDCC suppression factors,
- precompute scale-dependent factors involving k_{rel} ,
- avoid evaluating suppression factors at extremely small scales,
- ensure smooth transitions between linear and non-linear regimes.

The additional computational cost is modest.

B.9 Summary

This appendix provides a complete implementation strategy for RDCC-modified CMB lensing–galaxy cross-correlation physics in CLASS. The modifications are minimal, the numerical cost is low, and the resulting outputs are fully consistent with the analytical and semi-analytic results derived in the main text.

Appendix C: Forecast Tables for Planck, ACT, SPT, Euclid, LSST, and SKA

This appendix provides forecast tables for detecting the RDCC-modified $\kappa \times g$ signatures with Planck, ACT, SPT, Euclid, LSST, and SKA. We summarize the expected signal-to-noise ratios, detection significances, and constraints on the relaxation scale k_{rel} and the scale-dependent potential evolution.

C.1 Planck Forecasts

Observable	RDCC Signature	Planck Sensitivity
Large-scale $\kappa \times g$ enhancement	+10–20%	$> 3\sigma$
Small-scale suppression	–10–15%	$> 2\sigma$
Turnover in $C_\ell^{\kappa g}$	at ℓ_{rel}	$> 2\sigma$
Smoother lensing potential	detectable	$> 2\sigma$
Constraint on k_{rel}	—	$\pm 0.04 h \text{ Mpc}^{-1}$

Table 1: Planck forecast sensitivities to RDCC $\kappa \times g$ signatures.

C.2 ACT Forecasts

Observable	RDCC Signature	ACT Sensitivity
Small-scale suppression	–15–25%	$> 3\sigma$
Turnover in $C_\ell^{\kappa g}$	at ℓ_{rel}	$> 3\sigma$
Smoother convergence kernel	detectable	$> 3\sigma$
Constraint on k_{rel}	—	$\pm 0.03 h \text{ Mpc}^{-1}$

Table 2: ACT forecast sensitivities to RDCC $\kappa \times g$ signatures.

C.3 SPT Forecasts

Observable	RDCC Signature	SPT Sensitivity
Small-scale suppression	–15–25%	$> 3\sigma$
High- ℓ turnover	at ℓ_{rel}	$> 2\sigma$
Reduced potential decay	detectable	$> 3\sigma$
Constraint on k_{rel}	—	$\pm 0.03 h \text{ Mpc}^{-1}$

Table 3: SPT forecast sensitivities to RDCC $\kappa \times g$ signatures.

C.4 Euclid Forecasts

Observable	RDCC Signature	Euclid Sensitivity
Large-scale enhancement	+10–20%	$> 3\sigma$
Small-scale suppression	–20–30%	$> 4\sigma$
Turnover in $C_\ell^{\kappa g}$	at ℓ_{rel}	$> 3\sigma$
Redshift-dependent signatures	strong	$> 3\sigma$
Constraint on k_{rel}	—	$\pm 0.03 h \text{ Mpc}^{-1}$

Table 4: Euclid forecast sensitivities to RDCC $\kappa \times g$ signatures.

C.5 LSST Forecasts

Observable	RDCC Signature	LSST Sensitivity
Large-scale enhancement	+15–25%	$> 4\sigma$
Turnover in $C_\ell^{\kappa g}$	at ℓ_{rel}	$> 4\sigma$
Tracer-dependent amplitudes	strong	$> 3\sigma$
High- z sensitivity	excellent	$> 4\sigma$
Constraint on k_{rel}	—	$\pm 0.02 h \text{ Mpc}^{-1}$

Table 5: LSST forecast sensitivities to RDCC $\kappa \times g$ signatures.

C.6 SKA Forecasts

Observable	RDCC Signature	SKA Sensitivity
Large-scale enhancement	+20%	$> 5\sigma$
Small-scale suppression	−20–30%	$> 4\sigma$
Turnover in $C_\ell^{\kappa g}$	at ℓ_{rel}	$> 4\sigma$
High- z tomography	strong	$> 4\sigma$
Constraint on k_{rel}	—	$\pm 0.02 h \text{ Mpc}^{-1}$

Table 6: SKA forecast sensitivities to RDCC $\kappa \times g$ signatures.

C.7 Combined Constraints

Parameter	RDCC Prediction	Combined Sensitivity
Relaxation scale k_{rel}	turnover in $\kappa \times g$	$\pm 0.015 h \text{ Mpc}^{-1}$
Large-scale enhancement	+10–25%	$> 5\sigma$
Small-scale suppression	−15–30%	$> 5\sigma$
Turnover in $C_\ell^{\kappa g}$	at ℓ_{rel}	$> 5\sigma$
Scale-dependent potential evolution	detectable	$> 4\sigma$

Table 7: Combined constraints from Planck, ACT, SPT, Euclid, LSST, and SKA.

C.8 Summary

These forecast tables demonstrate that Planck, ACT, SPT, Euclid, LSST, and SKA provide strong observational sensitivity to the RDCC $\kappa \times g$ signatures:

- enhanced large-scale correlations,
- suppressed small-scale correlations,
- tracer-dependent amplitudes,
- a turnover at ℓ_{rel} ,
- scale-dependent potential evolution.

Together, these surveys can constrain the relaxation scale to $\Delta k_{\text{rel}} \sim 0.015 h \text{ Mpc}^{-1}$ and detect the RDCC potential- evolution signatures at high significance.

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